

Robustness of ML Models on Breast Cancer Data Under Data Degradation

Ege SAYGILARI
31213851

Dataset: Wisconsin Breast Cancer (sklearn) • 569 samples, 30 features, Binary Classification
Models: Logistic Regression | SVM | Random Forest | XGBoost

1 INTRODUCTION

AI and ML models are widely used in medical diagnostics. Real-world data often suffers from missing values and noise. This study systematically evaluates how four ML classifiers degrade under controlled data corruption on the Wisconsin Breast Cancer dataset, identifying critical thresholds and recommending robust model choices.

2 DATASET

Source: sklearn.datasets.load_breast_cancer
Samples: 569 | **Features:** 30 | **Classes:** Malignant / Benign
Split: 80% train / 20% test (stratified, seed=42)

3 METHODOLOGY

- Load dataset → stratified train/test split
- Inject missing data (0–40%) or Gaussian noise ($\sigma=0-2$)
- Impute (mean / median) → StandardScaler normalization
- Train: Logistic Regression, SVM, Random Forest, XGBoost
- Evaluate: Accuracy & F1-score per corruption level
- Compare strategies & identify performance thresholds

5 KEY FINDINGS

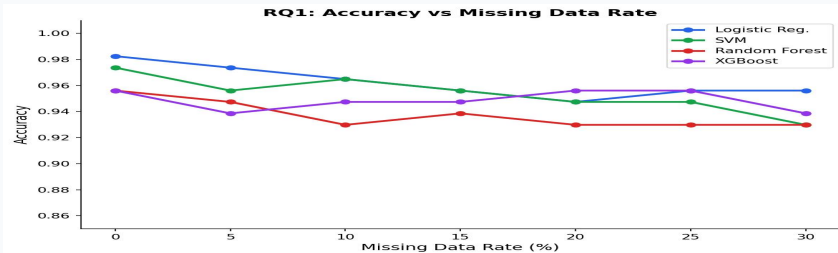
- RQ1:** All models degrade with missing data; tree-based models hold better
- RQ2:** RF & XGBoost most stable under noise; SVM drops fastest
- RQ3:** Mean vs median imputation: minimal difference across models
- RQ4:** Tree-based outperform linear on Accuracy & F1 at 20% degradation
- RQ5:** ~30–40% corruption: all models fall below 90% accuracy

7 CONCLUSION

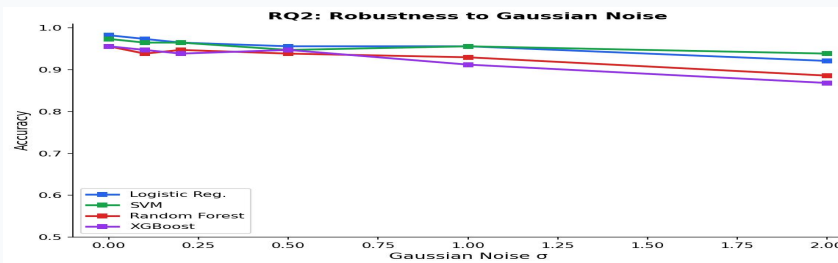
Tree-based models (RF, XGBoost) consistently outperform linear models. Imputation strategy has minimal impact. A ~30–40% corruption threshold is the critical failure point for all models.

4 RESEARCH QUESTIONS & RESULTS

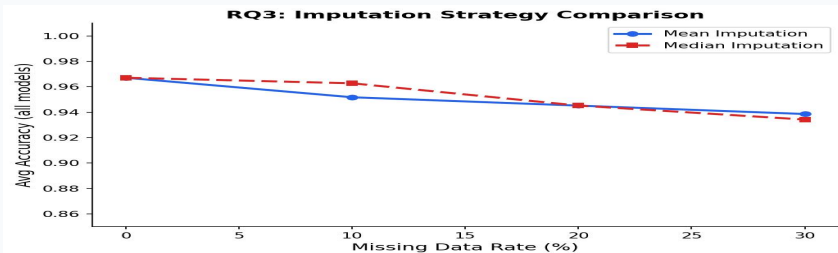
RQ1: Accuracy vs Missing Data Rate



RQ2: Robustness to Gaussian Noise



RQ3: Imputation Strategy Comparison

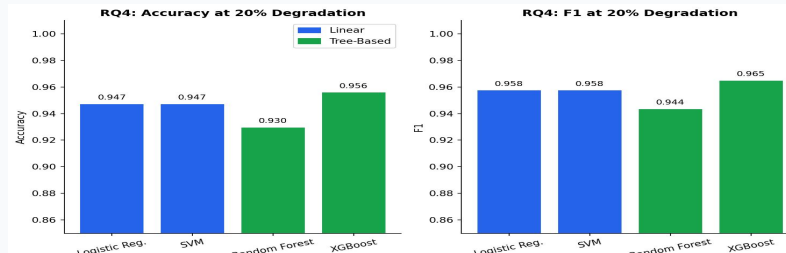


6 CONTRIBUTIONS

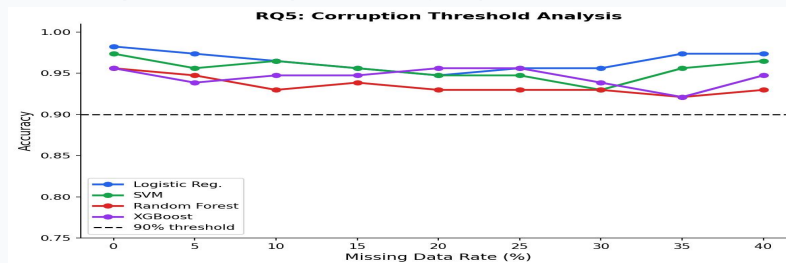
Systematic benchmark of 4 ML classifiers under realistic data degradation
Quantified corruption threshold (~30–40%) where all models fail clinically
Demonstrated imputation strategy has negligible effect on robustness
Provided actionable model recommendations for noisy medical datasets

4 RESULTS (cont.)

RQ4: Linear vs Tree-Based at 20% Degradation



RQ5: Corruption Threshold (90% accuracy line)



5 RQ SUMMARY TABLE

RQ	Question	Finding
RQ1	Missing data impact?	Tree-based models degrade slowest
RQ2	Gaussian noise robustness?	RF/XGBoost most stable
RQ3	Imputation strategy effect?	Mean ≈ Median (minimal diff.)
RQ4	Linear vs tree at 20%?	Tree-based win on Acc & F1
RQ5	Corruption threshold?	~30–40% breaks all models

6 FUTURE WORK

- Test on larger clinical datasets (TCGA, MIMIC)
- Explore deep learning robustness (MLP, CNN)
- Evaluate advanced imputation: KNN, MICE